

Jayachandra Galda

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PROFESSIONAL SUMMARY

AI Performance and Computer Vision Engineer with an M.S. in Artificial Intelligence and 1+ years of experience optimizing deep learning models and low-latency execution pipelines. Skilled in C++, Python, PyTorch, and execution profiling to isolate memory and compute bottlenecks. Cut inference latency 23% through multithreading, reduced memory overhead 48% via mixed-precision execution across vision and generative architectures..

EDUCATION

University at Buffalo, State University of New York

M.S. in Engineering Science (Artificial Intelligence)

Buffalo, NY

Jan 2025 – May 2026

Mahindra University

B.Tech in Computer Science and Engineering

Hyderabad, India

2020 – 2024

Certification: AWS Certified Data Engineer – Associate

TECHNICAL SKILLS

- **Parallel Programming & Acceleration:** CUDA C/C++, Multithreading, Concurrency, Dynamic Batching, Tensor-Level Operations, Latency Profiling, Memory Alignment, Cache Locality.
- **Languages & Core CS:** C++, C, Python, SQL, JavaScript; Data Structures & Algorithms, Numerical Computing, Object-Oriented Design, Design Patterns.
- **Model Performance:** Execution Profiling, Bottleneck Analysis, Mixed-Precision Execution, Model Compression, Memory Profiling, Edge Optimization.
- **Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, CNNs (YOLOv8, SqueezeNet), Variational Autoencoders (beta-VAE), LLM Inference (Llama-3.3-70B, Qwen2.5), OpenCV, scikit-learn, NumPy.
- **LLM & Retrieval Systems:** LangGraph, LangSmith, Ollama, Agentic Workflows, Hybrid Retrieval (BM25 + Dense), FAISS, Re-Ranking, Prompt Engineering, Groq API.
- **Backend & Cloud:** FastAPI, REST APIs, Async IO, AWS (EC2, S3, Lambda, ECS), PostgreSQL, GitHub Actions.
- **Engineering Practice:** System Design, Code Reviews, Agile/Scrum, pytest, CI/CD, Linux, Git, Docker, CMake.

WORK EXPERIENCE

QAS.AI

Buffalo, NY

AI & Automation Engineer

Jan 2026 – Jun 2026

- Built and shipped a low-latency RAG retrieval service and LLM inference API (FastAPI, Groq LLM, FAISS) serving real-time semantic search and ranking over large domain corpora.
- Designed its automated evaluation harness, scoring retrieval relevance and answer accuracy against curated reference sets so regressions were caught before release.
- Architected a multi-stage SQL ETL pipeline turning ~500K raw records into 1.6M rows of queryable time-series data.
- Engineered a vectorized decision engine (Python, NumPy, Pandas) evaluating 5 continuous data streams against a 7-step rule flow, replacing a manual review process.

University at Buffalo

Buffalo, NY

Graduate Research Assistant

Aug 2025 – Jan 2026

- Engineered *StrategyGen*, a multimodal ingestion pipeline converting raw ELAN annotation files into normalized event-level JSON via hybrid rule-based merging and a CPU-optimized SLM.
- Reduced processing turnaround by 60% (~20 hours/week) through batched tensor inference, memory-constrained CPU execution guardrails, and deterministic fallbacks.
- Scoped, containerized, and deployed the service end to end (Docker, Gradio, PyTorch); non-engineering collaborators now run it unaided.

Digitap.AI

Bengaluru, India

Machine Learning Engineer

Jan 2024 – Aug 2024

- Deployed a real-time detection pipeline (YOLOv8, OpenCV) processing high-throughput production video at 96.3% accuracy, with an input-validity filter that dropped corrupted frames.
- Cut end-to-end inference latency by 23% through C++ / Python optimization, multithreading, and dynamic batching, after profiling the pipeline to isolate the dominant bottlenecks.
- Engineered a lightweight SqueezeNet-based CNN for grayscale detection on low-compute edge hardware, trading model capacity against latency and memory budget while holding 98% accuracy.
- Developed a 3D facial alignment algorithm (X/Y/Z axes) that improved verification success rates by 21%, delivered on a shared release cadence with cross-functional code review.

PROJECTS

High-Performance Parallel Image Processing Engine

C++, CUDA, OpenCV, POSIX Threads

- Developed a zero-copy C++ frame preprocessing engine using POSIX multithreading and OpenCV matrix operations to process high-throughput video streams in parallel.
- Profiled matrix transformations and memory access patterns to optimize cache locality, cutting feature extraction latency 38% on resource-constrained hardware.

Unsupervised Latent Representation & Model Compression (beta-VAE)

PyTorch, Python

- Trained a beta-VAE (beta=4, 128-dim latent) on 6,800+ images to learn disentangled representations without labels; classification over frozen latents reached 98% accuracy and 0.997 ROC-AUC at a 95%+ dimensional compression ratio over raw input.

3D Cardiac DICOM Latent Compression (beta-VAE)

PyTorch, Mixed Precision, Memory Profiling

- Designing a 3D beta-VAE in PyTorch to compress multi-gigabyte volumetric cardiac data while preserving diagnostic structure.
- Profiled tensor memory allocation and applied mixed-precision execution, cutting memory overhead 48% under VRAM limits.